Similarly, the percentage of $\frac{3}{6}^{th}$ portion of work would be $\frac{3}{6}\times 100\% = \frac{300}{6}\% = 50\%$

Increase or decrease as Percentage

There are situations where we need to know the increase or decrease of a certain quantity as percentage. Let us see few examples.

Example 2.19 During Aadi sale the price of shirt decreased from ₹ 90 to ₹ 50. What is the percentage of decrease.

Solution

Original price = the price of the shirt before Aadi month

Amount of change = the decrease in the price = 90 – 50 = ₹ 40

$$\text{Therefore, the percentage of decrease} = \frac{\text{Amount of change}}{\text{Original amount}}\times 100$$

$$= \frac{40}{90}\times 100 = \frac{400}{9}$$

$$= 44\frac{4}{9}\%$$

Example 2.20 The number of literate persons in a city increased from 5 lakhs to 8 lakhs in 5 years. What is the percentage of increase?

Solution

Original amount = the number of literate persons initially = 5 lakhs

Amount of change = increase in the number of literate persons = 8 – 5 = 3 lakhs

$$\text{Therefore, the percentage of increase} = \frac{\text{Amount of change}}{\text{Original amount}}\times 100$$

$$= \frac{3}{5}\times 100 = 60\%$$

Try these

Level of water in a tank is increased from 35 litres to 50 litres in 2 minutes, what is the percentage of increase?

Profit or Loss as a Percentage

We have learnt already profit and loss of items. Now we will see how a profit or loss can be converted to percentage. That is, to find the profit % or loss %, we will see some examples.

Example 2.21 A shopkeeper bought a chair for ₹ 325 and sold it for ₹ 350. Find the profit percentage.

Solution

$$\text{Profit per cent} = \frac{\text{Profit}}{\text{C.P}}\times 100$$

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